

2010
**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total marks – 120

- Attempt Questions 1–8
- All questions are of equal value

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Total marks – 120

Attempt Questions 1–8

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (15 marks) Use a SEPARATE writing booklet.

(a) Find $\int \frac{x}{\sqrt{1+3x^2}} dx$. **2**

(b) Evaluate $\int_0^{\frac{\pi}{4}} \tan x \, dx$. **3**

(c) Find $\int \frac{1}{x(x^2+1)} dx$. **3**

(d) Using the substitution $t = \tan \frac{x}{2}$, or otherwise, evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{1+\sin x}$. **4**

(e) Find $\int \frac{dx}{1+\sqrt{x}}$. **3**

Question 2 (15 marks) Use a SEPARATE writing booklet.

(a) Let $z = 5 - i$.

(i) Find z^2 in the form $x + iy$. **1**

(ii) Find $z + 2\bar{z}$ in the form $x + iy$. **1**

(iii) Find $\frac{i}{z}$ in the form $x + iy$. **2**

(b) (i) Express $-\sqrt{3} - i$ in modulus–argument form. **2**

(ii) Show that $(-\sqrt{3} - i)^6$ is a real number. **2**

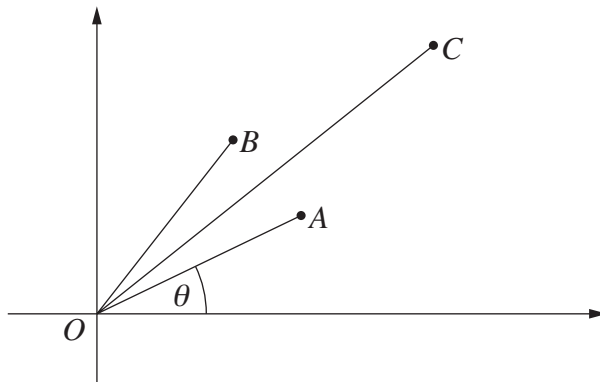
(c) Sketch the region in the complex plane where the inequalities $1 \leq |z| \leq 2$ and $0 \leq z + \bar{z} \leq 3$ hold simultaneously. **2**

Question 2 continues on page 5

Question 2 (continued)

- (d) Let $z = \cos \theta + i \sin \theta$ where $0 < \theta < \frac{\pi}{2}$.

On the Argand diagram the point A represents z , the point B represents z^2 and the point C represents $z + z^2$.



Copy or trace the diagram into your writing booklet.

- | | | |
|-------|--|---|
| (i) | Explain why the parallelogram $OACB$ is a rhombus. | 1 |
| (ii) | Show that $\arg(z + z^2) = \frac{3\theta}{2}$. | 1 |
| (iii) | Show that $ z + z^2 = 2\cos\frac{\theta}{2}$. | 2 |
| (iv) | By considering the real part of $z + z^2$, or otherwise, deduce that $\cos\theta + \cos 2\theta = 2\cos\frac{\theta}{2}\cos\frac{3\theta}{2}$. | 1 |

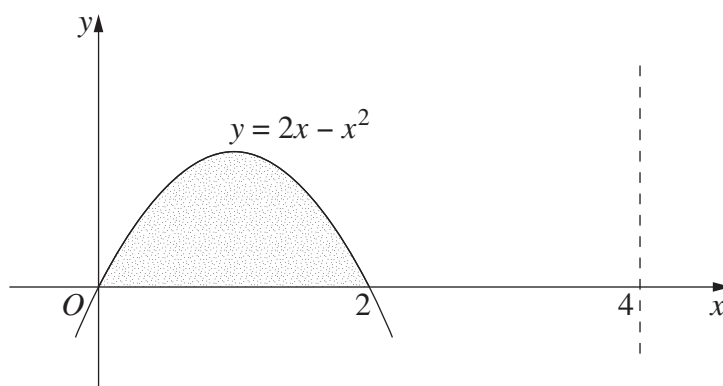
End of Question 2

Question 3 (15 marks) Use a SEPARATE writing booklet.

(a) (i) Sketch the graph $y = x^2 + 4x$. **1**

(ii) Sketch the graph $y = \frac{1}{x^2 + 4x}$. **2**

(b) The region shaded in the diagram is bounded by the x -axis and the curve $y = 2x - x^2$. **4**



The shaded region is rotated about the line $x = 4$.

Find the volume generated.

(c) Two identical biased coins are each more likely to land showing heads than showing tails. **2**

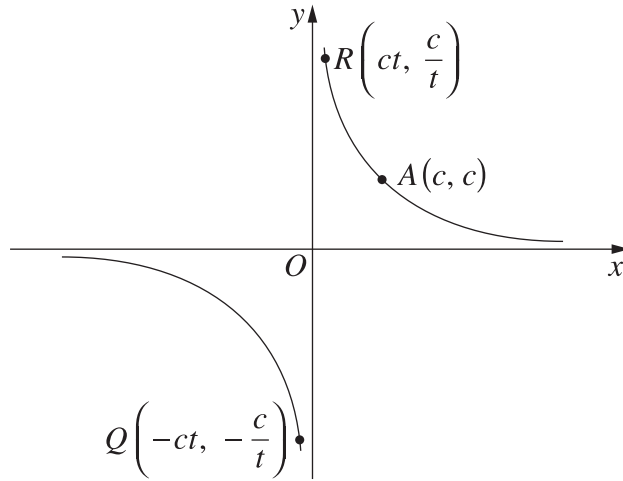
The two coins are tossed together, and the outcome is recorded. After a large number of trials it is observed that the probability that the two coins land showing a head and a tail is 0.48.

What is the probability that both coins land showing heads?

Question 3 continues on page 7

Question 3 (continued)

- (d) The diagram shows the rectangular hyperbola $xy = c^2$, with $c > 0$.



The points $A(c, c)$, $R\left(ct, \frac{c}{t}\right)$ and $Q\left(-ct, -\frac{c}{t}\right)$ are points on the hyperbola, with $t \neq \pm 1$.

- (i) The line ℓ_1 is the line through R perpendicular to QA . 2

Show that the equation of ℓ_1 is

$$y = -tx + c\left(t^2 + \frac{1}{t}\right).$$

- (ii) The line ℓ_2 is the line through Q perpendicular to RA . 1

Write down the equation of ℓ_2 .

- (iii) Let P be the point of intersection of the lines ℓ_1 and ℓ_2 . 2

Show that P is the point $\left(\frac{c}{t^2}, ct^2\right)$.

- (iv) Give a geometric description of the locus of P . 1

End of Question 3

Question 4 (15 marks) Use a SEPARATE writing booklet.

- (a) (i) A curve is defined implicitly by $\sqrt{x} + \sqrt{y} = 1$. 2

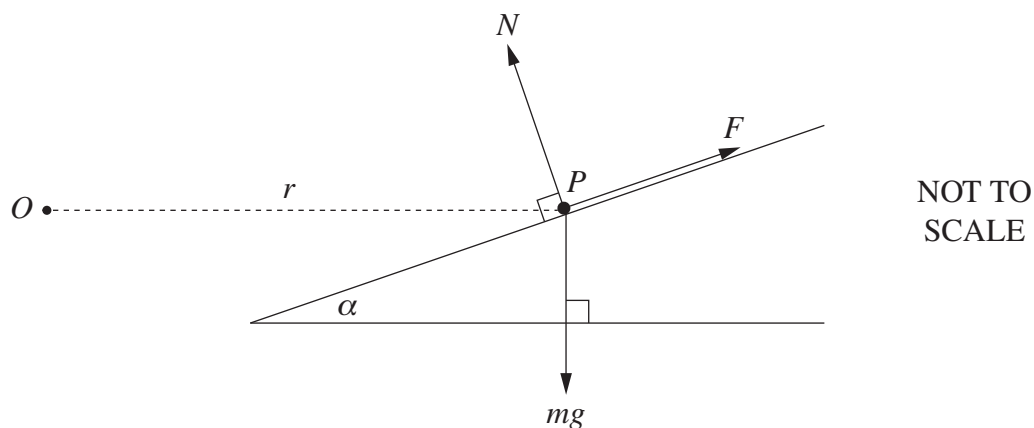
Use implicit differentiation to find $\frac{dy}{dx}$.

- (ii) Sketch the curve $\sqrt{x} + \sqrt{y} = 1$. 2

- (iii) Sketch the curve $\sqrt{|x|} + \sqrt{|y|} = 1$. 1

- (b) A bend in a highway is part of a circle of radius r , centre O . Around the bend the highway is banked at an angle α to the horizontal.

A car is travelling around the bend at a constant speed v . Assume that the car is represented by a point P of mass m . The forces acting on the car are a lateral force F , the gravitational force mg and a normal reaction N to the road, as shown in the diagram.



- (i) By resolving forces, show that $F = mg \sin \alpha - \frac{mv^2}{r} \cos \alpha$. 3
- (ii) Find an expression for v such that the lateral force F is zero. 1

Question 4 continues on page 9

Question 4 (continued)

- (c) Let k be a real number, $k \geq 4$.

Show that, for every positive real number b , there is a positive real number a **3**
such that $\frac{1}{a} + \frac{1}{b} = \frac{k}{a+b}$.

- (d) A group of 12 people is to be divided into discussion groups.

- (i) In how many ways can the discussion groups be formed if there are **1**
8 people in one group, and 4 people in another?
- (ii) In how many ways can the discussion groups be formed if there are **2**
3 groups containing 4 people each?

End of Question 4

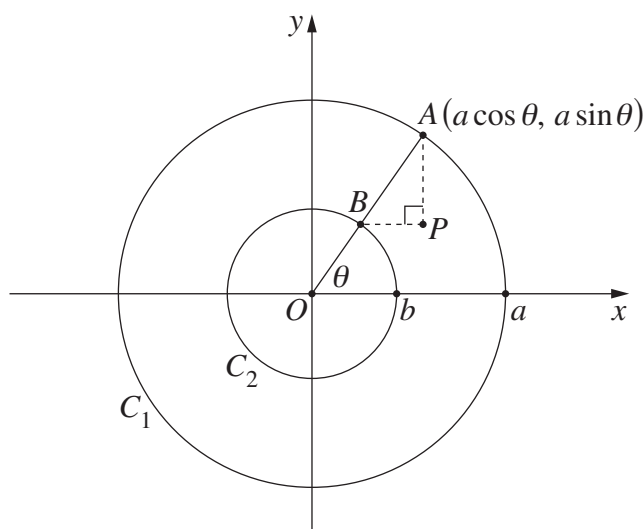
Question 5 (15 marks) Use a SEPARATE writing booklet.

- (a) The diagram shows two circles, C_1 and C_2 , centred at the origin with radii a and b , where $a > b$.

The point A lies on C_1 and has coordinates $(a \cos \theta, a \sin \theta)$.

The point B is the intersection of OA and C_2 .

The point P is the intersection of the horizontal line through B and the vertical line through A .



- | | |
|--|---|
| (i) Write down the coordinates of B . | 1 |
| (ii) Show that P lies on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. | 1 |
| (iii) Find the equation of the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at P . | 2 |
| (iv) Assume that A is not on the y -axis. | 2 |

Show that the tangent to the circle C_1 at A , and the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at P , intersect at a point on the x -axis.

Question 5 continues on page 11

Question 5 (continued)

(b) Show that

2

$$\int \frac{dy}{y(1-y)} = \ln\left(\frac{y}{1-y}\right) + c$$

for some constant c , where $0 < y < 1$.

(c) A TV channel has estimated that if it spends $\$x$ on advertising a particular program it will attract a proportion $y(x)$ of the potential audience for the program, where

$$\frac{dy}{dx} = ay(1-y)$$

and $a > 0$ is a given constant.

(i) Explain why $\frac{dy}{dx}$ has its maximum value when $y = \frac{1}{2}$.

1

(ii) Using part (b), or otherwise, deduce that

3

$$y(x) = \frac{1}{ke^{-ax} + 1}$$

for some constant $k > 0$.

(iii) The TV channel knows that if it spends no money on advertising the program then the audience will be one-tenth of the potential audience.

1

Find the value of the constant k referred to in part (c) (ii).

(iv) What feature of the graph $y = \frac{1}{ke^{-ax} + 1}$ is determined by the result in part (c) (i)?

1

(v) Sketch the graph $y = \frac{1}{ke^{-ax} + 1}$.

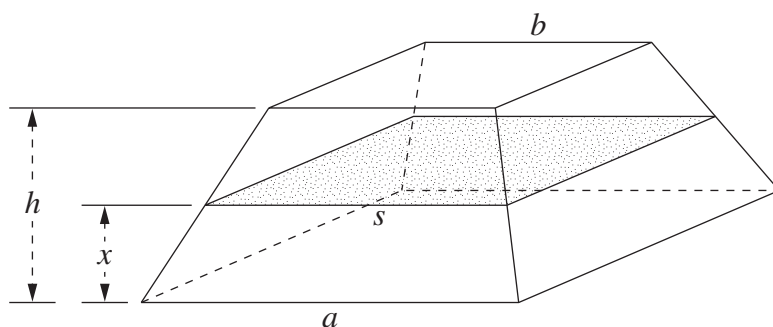
1

End of Question 5

Question 6 (15 marks) Use a SEPARATE writing booklet.

- (a) The diagram shows the frustum of a right square pyramid. (A frustum of a pyramid is a pyramid with its top cut off.)

The height of the frustum is h m. Its base is a square of side a m, and its top is a square of side b m (with $a > b > 0$).



A horizontal cross-section of the frustum, taken at height x m, is a square of side s m, shown shaded in the diagram.

(i) Show that $s = a - \frac{(a-b)}{h}x$. 2

(ii) Find the volume of the frustum. 2

- (b) A sequence a_n is defined by 3

$$a_n = 2a_{n-1} + a_{n-2},$$

for $n \geq 2$, with $a_0 = a_1 = 2$.

Use mathematical induction to prove that

$$a_n = (1 + \sqrt{2})^n + (1 - \sqrt{2})^n \text{ for all } n \geq 0.$$

Question 6 continues on page 13

Question 6 (continued)

(c) (i) Expand $(\cos \theta + i \sin \theta)^5$ using the binomial theorem. 1

(ii) Expand $(\cos \theta + i \sin \theta)^5$ using de Moivre's theorem, and hence show that 3

$$\sin 5\theta = 16\sin^5 \theta - 20\sin^3 \theta + 5\sin \theta.$$

(iii) Deduce that $x = \sin\left(\frac{\pi}{10}\right)$ is one of the solutions to 1

$$16x^5 - 20x^3 + 5x - 1 = 0.$$

(iv) Find the polynomial $p(x)$ such that $(x - 1)p(x) = 16x^5 - 20x^3 + 5x - 1$. 1

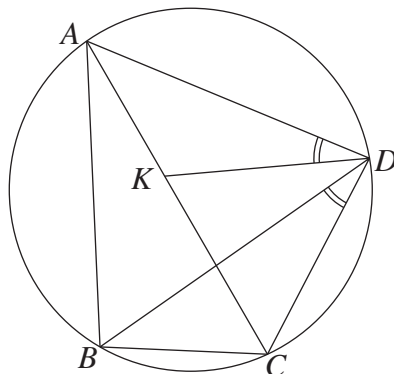
(v) Find the value of a such that $p(x) = (4x^2 + ax - 1)^2$. 1

(vi) Hence find an exact value for $\sin\left(\frac{\pi}{10}\right)$. 1

End of Question 6

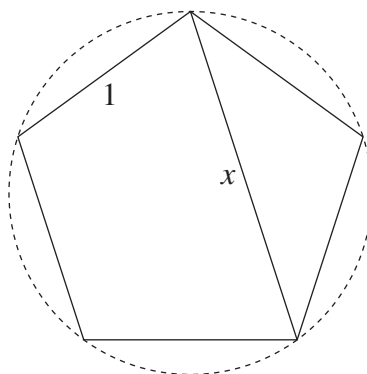
Question 7 (15 marks) Use a SEPARATE writing booklet.

- (a) In the diagram $ABCD$ is a cyclic quadrilateral. The point K is on AC such that $\angle ADK = \angle CDB$, and hence $\triangle ADK$ is similar to $\triangle BDC$.



Copy or trace the diagram into your writing booklet.

- (i) Show that $\triangle ADB$ is similar to $\triangle KDC$. 2
- (ii) Using the fact that $AC = AK + KC$, 2
show that $BD \times AC = AD \times BC + AB \times DC$.
- (iii) A regular pentagon of side length 1 is inscribed in a circle, as shown in the diagram. 2



Let x be the length of a chord in the pentagon.

Use the result in part (ii) to show that $x = \frac{1 + \sqrt{5}}{2}$.

Question 7 continues on page 15

Question 7 (continued)

- (b) The graphs of $y = 3x - 1$ and $y = 2^x$ intersect at $(1, 2)$ and at $(3, 8)$. 1

Using these graphs, or otherwise, show that $2^x \geq 3x - 1$ for $x \geq 3$.

- (c) Let $P(x) = (n - 1)x^n - nx^{n-1} + 1$, where n is an odd integer, $n \geq 3$.
- (i) Show that $P(x)$ has exactly two stationary points. 1
- (ii) Show that $P(x)$ has a double zero at $x = 1$. 1
- (iii) Use the graph $y = P(x)$ to explain why $P(x)$ has exactly one real zero other than 1. 2
- (iv) Let α be the real zero of $P(x)$ other than 1. 2
- Using part (b), or otherwise, show that $-1 < \alpha \leq -\frac{1}{2}$.
- (v) Deduce that each of the zeros of $4x^5 - 5x^4 + 1$ has modulus less than or equal to 1. 2

End of Question 7

Question 8 (15 marks) Use a SEPARATE writing booklet.

Let

$$A_n = \int_0^{\frac{\pi}{2}} \cos^{2n} x \, dx \quad \text{and} \quad B_n = \int_0^{\frac{\pi}{2}} x^2 \cos^{2n} x \, dx,$$

where n is an integer, $n \geq 0$. (Note that $A_n > 0$, $B_n > 0$.)

(a) Show that $n A_n = \frac{2n-1}{2} A_{n-1}$ for $n \geq 1$. 2

(b) Using integration by parts on A_n , or otherwise, show that 1

$$A_n = 2n \int_0^{\frac{\pi}{2}} x \sin x \cos^{2n-1} x \, dx \quad \text{for } n \geq 1.$$

(c) Use integration by parts on the integral in part (b) to show that 3

$$\frac{A_n}{n^2} = \frac{(2n-1)}{n} B_{n-1} - 2B_n \quad \text{for } n \geq 1.$$

(d) Use parts (a) and (c) to show that 1

$$\frac{1}{n^2} = 2 \left(\frac{B_{n-1}}{A_{n-1}} - \frac{B_n}{A_n} \right) \quad \text{for } n \geq 1.$$

(e) Show that $\sum_{k=1}^n \frac{1}{k^2} = \frac{\pi^2}{6} - 2 \frac{B_n}{A_n}$. 2

(f) Use the fact that $\sin x \geq \frac{2}{\pi} x$ for $0 \leq x \leq \frac{\pi}{2}$ to show that 1

$$B_n \leq \int_0^{\frac{\pi}{2}} x^2 \left(1 - \frac{4x^2}{\pi^2} \right)^n dx.$$

Question 8 continues on page 17

Question 8 (continued)

(g) Show that $\int_0^{\frac{\pi}{2}} x^2 \left(1 - \frac{4x^2}{\pi^2}\right)^n dx = \frac{\pi^2}{8(n+1)} \int_0^{\frac{\pi}{2}} \left(1 - \frac{4x^2}{\pi^2}\right)^{n+1} dx.$ 1

(h) From parts (f) and (g) it follows that 2

$$B_n \leq \frac{\pi^2}{8(n+1)} \int_0^{\frac{\pi}{2}} \left(1 - \frac{4x^2}{\pi^2}\right)^{n+1} dx.$$

Use the substitution $x = \frac{\pi}{2} \sin t$ in this inequality to show that

$$B_n \leq \frac{\pi^3}{16(n+1)} \int_0^{\frac{\pi}{2}} \cos^{2n+3} t dt \leq \frac{\pi^3}{16(n+1)} A_n.$$

(i) Use part (e) to deduce that 1

$$\frac{\pi^2}{6} - \frac{\pi^3}{8(n+1)} \leq \sum_{k=1}^n \frac{1}{k^2} < \frac{\pi^2}{6}.$$

(j) What is $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{k^2}$? 1

End of paper

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

2010 HSC Mathematics Extension 2 Marking Guidelines

Question 1 (a)

Criteria	Marks
• Correct solution	2
• Shows evidence of implicitly or explicitly substituting $u = 1 + 3x^2$	1

Question 1 (b)

Criteria	Marks
• Correct solution	3
• Obtains correct primitive	2
• Recognises $\tan x = \frac{\sin x}{\cos x}$	1

Question 1 (c)

Criteria	Marks
• Correct solution	3
• Correct partial fraction decomposition, or equivalent merit	2
• Attempts partial fraction decomposition	1

Question 1 (d)

Criteria	Marks
• Correct solution	4
• Obtains correct primitive, or equivalent merit	3
• Obtains $\int \frac{2}{(1+t)^2} dt$, or equivalent merit	2
• Correctly substitutes for dx in terms of t , or for $\sin x$ in terms of t , or equivalent merit	1

Question 1 (e)

Criteria	Marks
• Correct solution	3
• Makes substantial progress	2
• Makes an appropriate substitution	1

Question 2 (a) (i)

Criteria	Marks
• Correct answer	1

Question 2 (a) (ii)

Criteria	Marks
• Correct answer	1

Question 2 (a) (iii)

Criteria	Marks
• Correct solution	2
• Attempts to multiply by $\frac{5+i}{5+i}$	1

Question 2 (b) (i)

Criteria	Marks
• Correct solution	2
• Correct modulus OR correct argument	1

Question 2 (b) (ii)

Criteria	Marks
• Correct solution	2
• Obtains $2^6(\cos 7\pi + i \sin 7\pi)$, or equivalent merit	1

Question 2 (c)

Criteria	Marks
• Correct sketch	2
• Correctly sketches one of the regions	1

Question 2 (d) (i)

Criteria	Marks
• Shows that $OA = OB = 1$	1

Question 2 (d) (ii)

Criteria	Marks
• Correct solution	1

Question 2 (d) (iii)

Criteria	Marks
• Correct proof	2
• Makes some progress	1

Question 2 (d) (iv)

Criteria	Marks
• Correct deduction	1

Question 3 (a) (i)

Criteria	Marks
• Correct sketch	1

Question 3 (a) (ii)

Criteria	Marks
• Correct sketch	2
• Identifies the vertical asymptotes, or equivalent merit	1

Question 3 (b)

Criteria	Marks
• Correct solution	4
• Obtains a correct primitive	3
• Obtains a correct integral	2
• Makes progress towards obtaining a correct integral	1

Question 3 (c)

Criteria	Marks
• Correct solution	2
• Observes that $P(1 - P) = 0.24$ (where P = probability heads, or tails)	1

Question 3 (d) (i)

Criteria	Marks
• Correct solution	2
• Finds the gradient of QA correctly, or equivalent merit	1

Question 3 (d) (ii)

Criteria	Marks
• Correct answer	1

Question 3 (d) (iii)

Criteria	Marks
• Correct solution	2
• Attempts to solve simultaneously the equations for ℓ_1 and ℓ_2 , or equivalent merit	1

Question 3 (d) (iv)

Criteria	Marks
• Correct description (i.e. the top branch of the hyperbola)	1

Question 4 (a) (i)

Criteria	Marks
• Correct solution	2
• Attempts to differentiate implicitly	1

Question 4 (a) (ii)

Criteria	Marks
• Correct sketch vertical at (0, 1) and horizontal at (1, 0)	2
• Sketch showing some relevant features	1

Question 4 (a) (iii)

Criteria	Marks
• Correct sketch	1

Question 4 (b) (i)

Criteria	Marks
• Correct solution	3
• Resolves forces, and attempts to eliminate N	2
• Resolves forces horizontally or vertically, or equivalent merit	1

Question 4 (b) (ii)

Criteria	Marks
• Correct solution	1

Question 4 (c)

Criteria	Marks
• Correct proof	3
• Deduces that a solution exists when $k \geq 4$, or equivalent merit	2
• Obtains a relevant quadratic equation in a , or equivalent merit	1

Question 4 (d) (i)

Criteria	Marks
• Correct answer	1

Question 4 (d) (ii)

Criteria	Marks
• Correct solution	2
• Obtains $\begin{pmatrix} 12 \\ 4 \end{pmatrix} \times \begin{pmatrix} 8 \\ 4 \end{pmatrix}$, or equivalent merit	1

Question 5 (a) (i)

Criteria	Marks
• Correct answer	1

Question 5 (a) (ii)

Criteria	Marks
• Correct solution	1

Question 5 (a) (iii)

Criteria	Marks
• Correct equation	2
• Finds the gradient of the tangent, or equivalent merit	1

Question 5 (a) (iv)

Criteria	Marks
• Correct solution	2
• Finds where one of the tangents cuts the x axis, or equivalent merit	1

Question 5 (b)

Criteria	Marks
• Correct solution	2
• Attempts to differentiate $\ln\left(\frac{y}{1-y}\right)$, or attempts to use partial fractions on $\frac{1}{y(1-y)}$, or equivalent merit	1

Question 5 (c) (i)

Criteria	Marks
• Correct solution	1

Question 5 (c) (ii)

Criteria	Marks
• Correct solution	3
• Attempts to solve a correct equation for y	2
• Obtains $\ln\left(\frac{y}{1-y}\right) = ax + c$	1

Question 5 (c) (iii)

Criteria	Marks
• Correct answer	1

Question 5 (c) (iv)

Criteria	Marks
• Correct solution $\left(\text{ie inflection at } y = \frac{1}{2} \text{ or, graph is steepest when } y = \frac{1}{2}\right)$	1

Question 5 (c) (v)

Criteria	Marks
• Correct sketch	1

Question 6 (a) (i)

Criteria	Marks
• Correct solution	2
• Attempts to use appropriate similar triangles, or equivalent merit	1

Question 6 (a) (ii)

Criteria	Marks
• Correct solution	2
• Obtains volume = $\int_0^h \left(a - \frac{a-b}{h}x \right)^2 dx$, or equivalent merit	1

Question 6 (b)

Criteria	Marks
• Correct proof	3
• Makes substantial progress	2
• Checks that the formula holds for $n = 0$ and $n = 1$, or equivalent merit	1

Question 6 (c) (i)

Criteria	Marks
• Correct expansion	1

Question 6 (c) (ii)

Criteria	Marks
• Correct solution	3
• Correctly equates imaginary parts of the two expansions, or equivalent merit	2
• Correct use of de Moivre's theorem, or equivalent merit	1

Question 6 (c) (iii)

Criteria	Marks
• Correct deduction	1

Question 6 (c) (iv)

Criteria	Marks
• Correct solution	1

Question 6 (c) (v)

Criteria	Marks
• Correct solution	1

Question 6 (c) (vi)

Criteria	Marks
• Correct solution	1

Question 7 (a) (i)

Criteria	Marks
• Correct proof	2
• Identifies any pair of equal angles	1

Question 7 (a) (ii)

Criteria	Marks
• Correct solution	2
• Obtains a relevant ratio, or equivalent merit	1

Question 7 (a) (iii)

Criteria	Marks
• Correct solution	2
• Obtains lengths of sides and diagonals of an appropriate cyclic quadrilateral	1

Question 7 (b)

Criteria	Marks
• Correct graphs, or equivalent merit	1

Question 7 (c) (i)

Criteria	Marks
• Correct solution	1

Question 7 (c) (ii)

Criteria	Marks
• Correct solution	1

Question 7 (c) (iii)

Criteria	Marks
• Correct graph and correct explanation	2
• Finds that the curve $y = P(x)$ has a minimum at $(1, 0)$ and a maximum at $(0, 1)$, or recognises that $P(x) \rightarrow \infty$ as $x \rightarrow \infty$, and $P(x) \rightarrow -\infty$ as $x \rightarrow -\infty$, or equivalent merit	1

Question 7 (c) (iv)

Criteria	Marks
• Correct solution	2
• Shows that $P\left(-\frac{1}{2}\right) \geq 0$, or equivalent merit	1

Question 7 (c) (v)

Criteria	Marks
• Correct solution	2
• Recognises that the zeros are $1, 1, \alpha, \beta, \bar{\beta}$, or equivalent merit	1

Question 8 (a)

Criteria	Marks
• Correct solution	2
• Uses integration by parts appropriately	1

Question 8 (b)

Criteria	Marks
• Correct solution	1

Question 8 (c)

Criteria	Marks
• Correct solution	3
• Makes substantial progress	2
• Uses integration by parts to introduce a factor of x^2 into the integral	1

Question 8 (d)

Criteria	Marks
• Correct solution	1

Question 8 (e)

Criteria	Marks
• Correct solution	2
• Shows that $\sum \frac{1}{k^2} = 2 \left(\frac{B_0}{A_0} - \frac{B_n}{A_n} \right)$, or correctly calculates $\frac{B_0}{A_0}$, or equivalent merit	1

Question 8 (f)

Criteria	Marks
• Correct solution	1

Question 8 (g)

Criteria	Marks
• Correct solution	1

Question 8 (h)

Criteria	Marks
• Correct solution	2
• Substitutes $x = \frac{\pi}{2} \sin t$ correctly	1

Question 8 (i)

Criteria	Marks
• Correct solution	1

Question 8 (j)

Criteria	Marks
• Correct solution	1

Mathematics Extension 2

2010 HSC Examination Mapping Grid

Question	Marks	Content	Syllabus outcomes
1 (a)	2	4.1	E8
1 (b)	3	4.1	E8
1 (c)	3	4.1, 7.6	E8
1 (d)	4	4.1	E8
1 (e)	3	4.1	E8
2 (a) (i)	1	2.1	E3
2 (a) (ii)	1	2.1	E3
2 (a) (iii)	2	2.1	E3
2 (b) (i)	2	2.2	E3
2 (b) (ii)	2	2.4	E3
2 (c)	2	2.5	E3
2 (d) (i)	1	2.3	E3
2 (d) (ii)	1	2.3	E3
2 (d) (iii)	2	2.3	E3
2 (d) (iv)	1	2.3	E3
3 (a) (i)	1	1.1	E6
3 (a) (ii)	2	1.5	E6
3 (b)	4	5.1	E7
3 (c)	2	8.0	HE3, E9
3 (d) (i)	2	3.3	E4
3 (d) (ii)	1	3.3	E4
3 (d) (iii)	2	3.3	E4
3 (d) (iv)	1	3.3	E4, E9
4 (a) (i)	2	1.8	E6
4 (a) (ii)	2	1.8	E6, E9
4 (a) (iii)	1	1.3, 1.8	E6, E9
4 (b) (i)	3	6.3.4	E5
4 (b) (ii)	1	6.3.4	E5
4 (c)	3	8.0	E2, E9
4 (d) (i)	1	8.0	HE3, E9
4 (d) (ii)	2	8.0	HE3, E9
5 (a) (i)	1	3.1	E4
5 (a) (ii)	1	3.1	E4
5 (a) (iii)	2	3.1	E4
5 (a) (iv)	2	3.1	E4, E9
5 (b)	2	4.1	E2

Question	Marks	Content	Syllabus outcomes
5 (c) (i)	1	6.1	E2, E6
5 (c) (ii)	3	6.1	E2
5 (c) (iii)	1	6.1	E5, E9
5 (c) (iv)	1	6.1	E2, E6, E9, HE4
5 (c) (v)	1	1.1, 1.8, 1.5, 6.1	E6
6 (a) (i)	2	5.1	E2, E7
6 (a) (ii)	2	5.1	E7
6 (b)	3	8.2	E2, E9
6 (c) (i)	1	8.0	E3
6 (c) (ii)	3	2.4	E3
6 (c) (iii)	1	7.4	E4
6 (c) (iv)	1	7.4	E4
6 (c) (v)	1	7.4	E4
6 (c) (vi)	1	2.4, 7.4	E2, E4
7 (a) (i)	2	8.1	E2, E9
7 (a) (ii)	2	8.1	E2, E9
7 (a) (iii)	2	8.1	E2, E9
7 (b)	1	1.9	E2, E6
7 (c) (i)	1	7.2	E2, E4, E6
7 (c) (ii)	1	7.2	E2, E6
7 (c) (iii)	2	7.2, 7.3	E2, E4
7 (c) (iv)	2	7.2	E2, E4
7 (c) (v)	2	7.2, 7.5	E2, E4
8 (a)	2	4.1	E8
8 (b)	1	4.1	E8
8 (c)	3	4.1	E8
8 (d)	1	4.1	E2, E8, E9
8 (e)	2	4.1	E2, E8, E9
8 (f)	1	4.1	E2, E8, E9
8 (g)	1	4.1	E2, E8, E9
8 (h)	2	4.1	E2, E8, E9
8 (i)	1	4.1, 8.0	E2, E8, E9
8 (j)	1	4.1, 8.0	E2, E8, E9